

torch.topk#

<https://pytorch.org/docs/stable/generated/torch.topk.html#torch.topk>

Description:

Returns the k largest elements of the given input tensor along a given dimension. If dim is not given, the last dimension of the input is chosen. If largest is False then the k smallest elements are returned. A namedtuple of (values, indices) is returned with the values and indices of the largest k elements of each row of the input tensor in the given dimension dim. The boolean option sorted if True, will make sure that the returned k elements are themselves sorted

Function Signature

```
torch.topk(input, k, dim=None, largest=True, sorted=True, *,  
out=None)#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (tuple, optional) – the output tuple of (Tensor, LongTensor) that can be optionally given to be used as output buffers

Code Examples

```
>>> x = torch.arange(1., 6.)
>>> x
tensor([ 1.,  2.,  3.,  4.,  5.])
>>> torch.topk(x, 3)
torch.return_types.topk(values=tensor([5., 4., 3.]),
indices=tensor([4, 3, 2]))
```

Notes & Warnings

NOTE When using `torch.topk`, the indices of tied elements are not guaranteed to be stable and may vary across different invocations.

Detailed Documentation

Documentation

Returns the k largest elements of the given input tensor along a given dimension.

If dim is not given, the last dimension of the input is chosen.

If largest is False then the k smallest elements are returned.

A namedtuple of (values, indices) is returned with the values and indices of the largest k elements of each row of the input tensor in the given dimension dim.

The boolean option sorted if True, will make sure that the returned k elements are themselves sorted

When using torch.topk, the indices of tied elements are not guaranteed to be stable and may vary across different invocations.

input (Tensor) – the input tensor.

k (int) – the k in “top-k”

dim (int, optional) – the dimension to sort along

largest (bool, optional) – controls whether to return largest or smallest elements

sorted (bool, optional) – controls whether to return the elements in sorted order

out (tuple, optional) – the output tuple of (Tensor, LongTensor) that can be optionally given to be used as output buffers

